An Efficient and Robust Resource Allocation Method for Quantum Key Distribution Networks

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***Abstract*—Quantum Key Distribution (QKD) technology leverages its inherent security advantages to ensure information-theoretic security for data transmission in networks. However, existing QKD networks still face critical challenges, including network congestion that stems from limited key resources and uneven resource allocation methods. Thus, in this paper, we focus on the issue of network congestion caused by bottleneck links and aim to achieve load balancing. Considering the limited key resources, we first introduce the key resource utilization ratio as an indicator of bottleneck links and formulate the resource allocation problem as an Integer Linear Programming (ILP) problem. To deal with the complexity of the ILP problem, especially in large-scale network scenarios, we design a heuristic algorithm that can obtain near-optimal solutions within polynomial time. Finally, we implement the proposed key resource allocation scheme in various real-world network topologies using a full-stack quantum network simulator. Compared to the existing algorithms, extensive results show that our method can reduce key resource consumption by up to 50% on bottleneck links and improve the robustness of QKD networks when facing burst quantum key agreement requests.**

***Index Terms*—QKD network, key resource allocation, load balance, key utilization ratio.**

I. Introduction

With the rapid development of quantum information technology, classical cryptographic algorithms based on the security assumption of computational complexity, such as Rivest-Shamir-Adleman (RSA) [1], face significant security threats posed by powerful quantum computers [2]. As a promising solution, Quantum Key Distribution (QKD) technology can achieve key distribution between two distant parties with information-theoretic security [3], [4], thus attracting increasing attention from both academic and industrial communities. To bring QKD technology from laboratory experiments to practical applications, existing studies have attempted to construct QKD networks and have achieved significant progress in recent years [5], [6]. For instance, the SECOQC QKD network [7], the Tokyo QKD network [8], and an integrated space-to-ground QKD network [9] have been built. It can be seen that a large-scale QKD network can be built to implement long-distance quantum key agreements between QKD nodes, thus supporting various security applications in the future [10], [11].

However, despite advancements in QKD networks, congestion caused by bottleneck links remains a significant challenge for the practical application of QKD technology. This issue is critical because network congestion increases request service delays and reduces throughput, thereby decreasing the efficiency and robustness of the QKD network. There are two main reasons for bottleneck links and network congestion. On the one hand, the limited key resources, i.e., the quantum keys generated by QKD devices for secure communication, result in insufficient key supply, making it difficult to meet numerous request requirements [12], [13], [14], [15]. The key generation rate of existing QKD devices generally decreases exponentially with the length of the quantum channel [16]. For example, the key generation rate is 22.2 Mbps over a 50 km fiber, and only 233 bps over a 328 km fiber [17]. These limited key generation rates make it difficult to support on-demand encryption services, especially when compared to the relatively high encryption requirements of various applications like 40 Mbps for video streaming [18], which exacerbates the challenge. On the other hand, uneven resource allocation methods lead to rapid depletion of available resources, making load balancing difficult to achieve. Traditional resource allocation methods, such as the shortest-path-first algorithm, tend to serve requests on certain paths, quickly exhausting available resources and creating bottleneck links. Meanwhile, key resources on less frequently used paths remain underutilized, which could provide relief to the congested links [19]. These uneven resource allocation methods prevent effective load balancing and further exacerbate network congestion.

To tackle the challenges of congestion in QKD networks, existing studies can be broadly classified into two categories.

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The first category focuses on improving the network's service capability by optimizing channel resource allocation in QKD integrated with classical telecommunication networks. For example, Cao et al. [20] proposed a scheme that combines Wavelength Division Multiplexing (WDM) and Time Division Multiplexing (TDM), allocating multiple time slots for quantum channels. Moghaddam et al. [21] explored the performance of QKD over optical networks using multi-core fibers and addressed the resource allocation problem for both quantum and classical channels. Halder et al. [22] introduced a resource allocation method for spectrally-spatially elastic optical networks, optimizing spectrum allocation while minimizing interference between quantum and classical channels. More recently, Ren et al. [23] applied Space Division Multiplexing (SDM) to enable the coexistence of quantum and classical channels in the same fiber, thus increasing the throughput of QKD networks. Although these methods enhance the network's service capabilities, challenges remain in providing on-demand services due to the inherent limitations of current QKD devices.

The second category of research focuses on characterizing quantum key agreement requests and allocating key resources more efficiently [24]. Mehic et al. [25] proposed a quality-of-service-based routing protocol aimed at achieving high scalability and minimizing key resource consumption. However, this method is designed for decentralized QKD networks and cannot efficiently allocate key resources across the entire network. As a result, it may lead to key resource shortages, bottleneck links, and network congestion. Recognizing the importance of maintaining the quality of service in QKD networks while managing cost constraints, Li et al. [26] developed a flow-based mathematical model to describe a QKD network. Although this model highlights the impact of bottleneck links, it does not address the issue of efficiently allocating key resources across the network to mitigate the occurrence of bottleneck links. Zhou et al. [27] acknowledged the critical issue of limited quantum key resources and designed a key management and data scheduling scheme to optimize the utility of data transmission. However, the use of Lyapunov optimization in their approach imposes artificial limitations on key storage and does not fully prevent the emergence of bottleneck links within QKD networks. In summary, existing methods still lack consideration of network congestion and bottleneck links from the perspective of resource allocation, and more efficient solutions are needed to address these challenges.

To address these challenges and improve the resource utilization efficiency and service capacity of existing QKD networks, we propose an efficient and robust resource allocation method. First, we establish a mathematical model based on network flow to describe the on-demand service problem in QKD networks. We then introduce the key utilization ratio as a metric to identify bottleneck links in the network and formulate an Integer Linear Programming (ILP) model to optimize resource allocation. To solve this ILP problem, we first obtain the optimal solution for small-scale networks using a solver. To enable scalability for large-scale networks, we then design a heuristic algorithm that can solve the problem in polynomial time complexity, significantly enhancing the practical applicability of the solution. To demonstrate the advantages of our method, we adopt SimQN[1] [28], a discrete-event simulation platform to compare the proposed method with existing resource allocation methods. Extensive results show that our method can effectively handle burst quantum key agreement requests and reduce the ratio of blocked requests. The main contributions of this paper can be summarized as follows:

- To address the congestion problem caused by bottleneck links in QKD networks, we propose an efficient and robust resource allocation method to optimize key resource usage and improve the network's service capability. Considering the limitations of current QKD networks, such as limited key generation rates and the risk of bottleneck links, we formulate the resource allocation problem as an ILP model to achieve load balancing and prevent resource shortages.
- To efficiently solve the resource allocation problem, we propose a heuristic algorithm based on key utilization ratios, which identifies bottleneck links and allocates resources in a load-balanced manner. On the one hand, compared to traditional ILP methods with non-polynomial time complexity, this heuristic algorithm significantly reduces running time. On the other hand, the adaptability of the proposed heuristic algorithm makes it not only suitable for metropolitan networks but also scalable to large-scale long-haul backbone networks.
- To evaluate the performance of the proposed method, extensive simulations are conducted in various realistic network topologies using a system-level quantum network simulation platform. Simulation results demonstrate that our method can significantly reduce key resource consumption on bottleneck links by up to 50% compared to existing algorithms and improve the robustness of QKD networks when facing burst key agreement requests.

The rest of this paper is organized as follows. We first introduce the background of QKD networks and our motivation in Section II. Section III states the problem we are studying in QKD networks and formulates the resource allocation problem as an ILP problem. Furthermore, we present the heuristic algorithm design in Section IV. We perform large-scale simulations with the multiple real-world topologies on SimQN and compare the proposed method with existing resource allocation methods in Section V. Finally, we conclude and discuss this work in Section VI.

II. Background and Motivation

In this section, we first introduce the QKD networks and quantum key agreement between distant nodes in Section II-A. Following this, we analyze the existing issues related to resource allocation within current QKD networks in Section II-B and propose corresponding design goals in Section II-C.

[1]The simulation platform we used in this study is an open-source project our team developed on GitHub: https://github.com/ertuil/simqn.

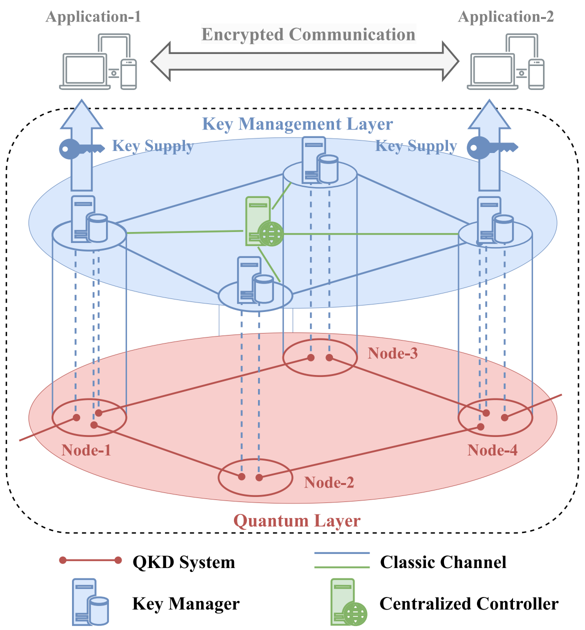
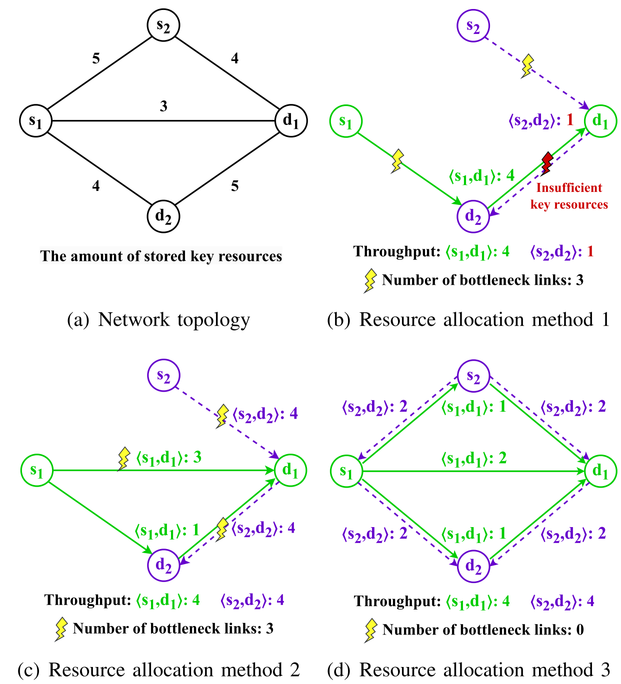


Fig. 1. The layered architecture of a QKD network.



(a) Network topology (b) Resource allocation method 1

(c) Resource allocation method 2 (d) Resource allocation method 3

Fig. 2. Motivation examples. The solid line and dashed line represent the service paths for request pair $\langle s_1, d_1\rangle$ and request pair $\langle s_2, d_2\rangle$, respectively. The numbers indicate the resource consumption on each edge while serving the corresponding request. The lightning symbol denotes resource depletion, indicating a bottleneck link.

A. Background

In a QKD network, as shown in Fig. 1, each QKD node consists of multiple QKD devices and a key manager. The QKD system can use point-to-point QKD protocols, such as BB84 protocol [29], to distribute quantum keys between adjacent QKD nodes. The key manager stores keys generated by QKD systems and provides on-demand key supply services to cryptographic applications upon receiving quantum key agreement requests. Each node in the QKD network is fully trusted [30], [31], installing the same components, e.g., computing unit and QKD transceiver. Each link of the QKD system contains a quantum channel and an authenticated classical channel. The quantum channel is used to transmit quantum signals, and the classical channel supports the secure interaction of classical information between QKD nodes during post-processing.

According to the standardized design specification for QKD network architectures [32], key relaying must be controlled by a centralized controller. This is required whenever there is a quantum key agreement request between remote nodes. First, the centralized controller periodically collects both quantum key agreement requests and key resource information from the entire network and then makes unified decisions based on them. The information on quantum key agreement requests contains the number of key packets requested, the source node, and the destination node. After that, a source node, under the guidance of the centralized controller, transmits the raw key materials to the destination node by relaying the requested key packets hop by hop along a selected path consisting of one or more QKD nodes. To ensure the security of the key relay process, the key packets from the source node must be protected at each hop using the OTP encryption algorithm. Once the destination node receives the key packets, it can use the received key to establish end-to-end secure communication with the source node. In this case, the time consumed by the secret-key request includes the time for transmitting keys over a link, i.e., the transmission delay, as well as the processing delay for encrypting and decrypting the transmitted key at each QKD relay. Furthermore, the traffic load is the total key resource consumption on the edges during the end-to-end secure communication, which is determined by both the amount of requested keys and the number of requests.

B. Motivation

In a real QKD network, compared to relatively high encryption requirements from various applications, e.g., 40 Mbps for video streaming [18], the key generation rate of QKD links are limited, e.g., 22.2 Mbps over a 50 km fiber [17]. Thus, the key resources in a QKD network are precious. Fig. 2 shows our motivation, highlighting how uneven resource allocation can lead to bottleneck links within a QKD network with four nodes.

As shown in Fig. 2(a), there are two quantum key agreement request pairs, $\langle s_1, d_1\rangle$ and $\langle s_2, d_2\rangle$, each request requires 4 units of key resources. When the resource allocation method shown in Fig. 2(b) is applied, both requests are routed through the link (d_1, d_2). However, due to insufficient key resources on (d_1, d_2), it is difficult to provide the on-demand service for both quantum key agreement requests, leading to a decrease in network throughput. Moreover, after serving two requests, there are no surplus key resources on the adjacent links of node d_2, making d_2 impossible to securely communicate with other nodes, thus reducing the robustness of the QKD network.

The same problem also exists in the second scheme, as shown in Fig. 2(c). Although selecting multiple paths for load balance can improve network throughput, bottleneck links still occur, preventing node d_1 from securely communicating with other nodes. In contrast, some links within the QKD network remain idle for extended periods under this resource allocation scheme. The key resources on these idle links accumulate over time, which reduces the overall utilization of key resources in the QKD network. Additionally, storing keys in key managers for too long a time will reduce the security of quantum keys [33]. The keys that have been stored for more than a certain period of time must be discarded or refreshed by consuming other keys with high-level security [34], contributing to the waste of valuable key resources.

In a real QKD network, the highly dynamic nature of requests makes it challenging to efficiently utilize the key resources within the QKD network. To make full use of the key resources, multiple optional paths can be used in resource allocation to satisfy secret-key requests, thereby achieving load balance. As illustrated in Fig. 2(d), the request pair $\langle s_1, d_1 \rangle$ and $\langle s_2, d_2 \rangle$ can choose three paths ($s_1 - d_1$, $s_1 - s_2 - d_1$, and $s_1 - d_2 - d_1$) and two paths ($s_2 - s_1 - d_2$ and $s_2 - d_1 - d_2$) to establish end-to-end secure communication, respectively. This resource allocation scheme not only ensures that the network can provide on-demand services for multiple users but also prevents the generation of bottleneck links.

C. Design Goal

To address the challenges posed by the dynamic nature of quantum key agreement requests and the limited key resources in QKD networks, we propose the following design goals.

- *High request service ratio:* In a QKD network, due to the randomness of requests and the limited key resources, network congestion can lead to service delays and a decrease in throughput. Therefore, the QKD network can make full use of the available key resources across the entire network to meet as many requests as possible, thereby achieving a high request service ratio.
- *Load balancing:* Uneven resource allocation methods lead to the generation of bottleneck links, reducing both the QKD network efficiency and robustness. Therefore, when a QKD network provides services to requests, it is crucial to balance the load on different paths to prevent congestion caused by bottleneck links.

By achieving these design goals, the QKD network can make full use of the limited key resources to serve dynamic quantum key agreement requests while preventing bottleneck links and network congestion, thereby enabling efficient and robust resource allocation.

III. System Model and Problem Formulation

A. System Model

In a typical centralized QKD network $G(V, E)$, each node is equipped with identical QKD transmitters and receivers and is connected to neighboring nodes via both quantum and classical channels. However, due to different optical fiber lengths between adjacent nodes, the key generation rates for different links differ. The generated quantum keys, serving as the key resources in the network, are transmitted to a key manager for storage and future use. The QKD network makes decisions and provides on-demand services for requests based on time slots. The traffic load is the total key resource consumption on the edges when providing services for requests.

TABLE I
Notations and Definitions

Notations	Definitions
$G(V,E)$	QKD network topology.
V	The set of nodes in the QKD network.
E	The set of edges in the QKD network.
R	The set of requests in the QKD network.
r_i	The amount of key resources demanded by request i.
s_i	The source node of the request i.
d_i	The destination node of the request i.
F	The set of allocated key resources in the QKD network.
$f_i(u,v)$	The amount of key resources consumed on edge (u,v) when request i relays keys between node u and node v.
$C(u,v)$	Amount of key resources stored on edge (u,v).
$g(u,v)$	The average key generation rate of edge (u,v).
$\alpha(u,v)$	The key utilization ratio on edge (u,v).
α	Maximum key utilization ratio in the QKD network.
$t_i(u,v)$	The transmission delay of request i on edge (u,v).
$T_i(u)$	Processing delay consumed by the encryption and decryption process on intermediate node u.
T_d	The decision periods in the QKD network.
D	The length of time slots in the QKD network.
P_i	The set of optional paths for request i.

Specifically, during each time slot (T, $T + D$], the centralized controller collects the set of requests R from the entire network. Based on the available key resources stored on the edge (u, v), denoted as $C(u,v)$, the controller determines the set of routing paths P and the corresponding allocated quantum key resources F. Starting from the source node, each node along the selected path relays the end-to-end key hop-by-hop to the destination node. The upper-layer request utilizes this end-to-end key to encrypt application data, achieving quantum-secure communication. After the requests in time slot (T, $T + D$] are serviced, the newly generated quantum keys are added to the key manager, and the stored key resources are updated as:

$$C'(u,v) = C(u,v) - \sum_i f_i(u,v) + g(u,v) \cdot D, \qquad (1)$$

where $C'(u,v)$ is the updated key resource for edge (u, v), $\sum_i f_i(u,v)$ represents the total consumed quantum key resources, and $g(u,v)$ denotes the average key generation rate of edge (u, v) in time slot (T,$T + D$]. The notations used in this paper are listed and defined in Table I.

B. Problem Definition and Constraints

Based on the analysis in Section II-B, when providing on-demand service for requests in the network, it is necessary to consider the potential bottleneck links to achieve load balancing. To evaluate the consumption of key resources on each link, we define a key utilization ratio $\alpha(u,v)$, which is the ratio of the total key resources consumed on a given link to the available key resources on that link during a time slot.

Specifically, the key utilization ratio is calculated as:

$$\alpha(u,v) = \frac{\sum_i f_i(u,v)}{C(u,v)}, \forall (u,v) \in E, \tag{2}$$

where $\sum_i f_i(u,v)$ represents the total key resources consumed on edge (u, v), which also corresponds to the traffic load on edge (u, v). If $\alpha(u,v) = 1$, this indicates that the edge (u, v) has reached its maximum capacity, becoming a bottleneck link. A smaller key utilization ratio indicates a lighter load on the link. When $\alpha(u,v) = 0$, it indicates that the link is idle, with no key resources being consumed. Similarly, according to Eq. (2), the maximum key utilization ratio in the QKD network can be calculated as

$$\alpha = \max_{f_i(u,v)} \alpha(u,v). \tag{3}$$

When $\alpha = 1$, it means that at least one bottleneck link exists in the QKD network. This occurs when the key resource consumption on a particular edge reaches its maximum capacity, meaning that the link is fully utilized, and no additional key resources can be transmitted along that path. In such cases, the QKD network may experience congestion, leading to request service delays, inefficient resource utilization, and potentially a failure to meet the demand for on-demand services. A smaller maximum key utilization ratio α indicates that the network's key resources are being used more efficiently, with fewer links reaching their maximum capacity. This leads to better overall network performance, as resources are not concentrated on a few critical links, thus reducing the risk of congestion and bottleneck links. Therefore, our objective is to minimize the maximum key utilization ratio across all edges, thus achieving load balancing in the QKD network.

The resource allocation problem for achieving load balancing can be described as follows: Given a QKD network $G(V, E)$, finding resource allocation methods, including the set of routing paths P and the corresponding quantum resource allocation F, to minimize the maximum key utilization ratio α. Furthermore, the resource allocation problem can be formally defined as the following optimization problem:

$$\min \quad \max_{f_i(u,v)} \quad \alpha(u,v) \tag{4}$$

subject to:

$$\sum_u [f_i(s_i,u) - f_i(u,s_i)] = r_i, \forall s_i \in V, \tag{4a}$$

$$\sum_u [f_i(u,d_i) - f_i(d_i,u)] = r_i, \forall d_i \in V, \tag{4b}$$

$$\sum_u [f_i(u,v) - f_i(v,u)] = 0, \forall v \in V - \{s_i, d_i\}, \tag{4c}$$

$$\sum_i [f_i(u,v) + f_i(v,u)] \leq C(u,v), \forall (u,v) \in E, \tag{4d}$$

$$T_d + \sum_{(u,v)} t_i(u,v) + \sum_u T_i(u) \leq D, \forall i \in R, \tag{4e}$$

$$r_i \in N, C(u,v) \in N, f_i(u,v) \in N. \tag{4f}$$

Constraint (4a) ensures that the source node s_i sends out the required amount of key resources r_i for each request i. Constraint (4b) ensures that the destination node d_i receives the required amount of key resources r_i for each request i. Constraint (4c) ensures that any intermediate node v properly forwards key packets between nodes. Constraints (4a) to (4c) ensure that the QKD network makes full use of the available key resources to provide on-demand services for the request i in the request set R. Constraint (4d) ensures that the total quantum keys consumed on each edge do not exceed the available key resources.

To provide on-demand services to users within a given time slot, it is crucial to ensure that the total time consumed by the actual relay process is bounded by the time slot's duration. The process begins with the centralized controller making resource allocation decisions, which are then sent to each QKD node during the decision period T_d. Once the decisions are made, the relay process for the quantum key agreement begins. In this process, the total time consumed in this process includes the total transmission delay $\sum_{(u,v)} t_i(u,v)$ and the total processing delay $\sum_u T_i(u)$ along the selected path. Since the decision information is only valid within the current time slot, it is critical that the entire service process is completed within the current time slot. Constraint (4e) ensures that the total time for key agreement, including both transmission and processing, does not exceed the time slot D, thereby ensuring that the service is provided within the time limits. Furthermore, because the service must be completed within the current time slot, this time consumption constraint also prevents the selection of excessively long paths. This serves as a mechanism to balance resource consumption with load-balancing effects, ensuring that the network can effectively distribute load without overusing key resources.

We use the maximum key utilization ratio of the QKD network as the optimization objective. By minimizing the maximum key resource utilization ratio, we aim to avoid the generation of bottleneck links and achieve load balance in the QKD network. Based on the constraints (4a) to (4f), we can formulate an ILP problem. Solving this ILP problem will yield resource allocation results in the QKD network. However, the ILP problem is NP-complete [35]. Although solving the ILP problem can yield the optimal solution, the time complexity of solving the ILP problem is high, making it infeasible for large networks. Therefore, we propose the Key Utilization Ratio based (KUR-based) algorithm.

IV. Heuristic Algorithm Design

In this section, we first analyze the existing solution for the proposed ILP problem. However, it is not scalable for large-scale QKD networks owing to its high computational complexity. To address this limitation, we propose the KUR-based algorithm, which provides a more efficient solution to the original ILP problem. After that, we analyze the challenges posed by inevitable bottleneck links in large-scale QKD networks, such as long-haul backbone networks. To address this challenge, we further propose the Long-haul Backbone Network KUR-based algorithm to make it suitable for large-scale QKD networks.

Algorithm 1: KUR-Based Algorithm

Input: $G = (V,E),R,D$;
Output: Assigned $f_i(u,v)$ in terms of allocated links, α, updated stored key resources;

1 Initialize $\alpha \leftarrow 0$;
2 **for** *each secret-key request* $i \in R$ **do**
3   Find the path set P_i with time consumption constraint by Dijkstra's algorithm;
4   **for** *each key packet* r_i^j **do**
5   **for** *each path* $p_i^j \in P_i$ **do**
6   **for** *each edge* $(u,v) \in p_i^j$ **do**
7   **if** $C(u,v) \leq \sum_i f_i(u,v) + r_i^j$ **then**
8           $f_i^j(u,v) \leftarrow f_i(u,v) + r_i^j$;
9   Update $\alpha^j(u,v)$ based on $f_i^j(u,v)$;
10   **end**
11   **end**
12       $\alpha^j \leftarrow \max_{f_i(u,v)} \alpha^j(u,v)$;
13   **end**
14     $\alpha \leftarrow \min \alpha^j$;
15     $f_i(u,v) \leftarrow f_i^j(u,v)$;
16   **end**
17 **end**

A. KUR-Based Algorithm

There are conventional heuristic techniques for solving ILP problems, such as the branch and bound method, which can yield the optimal solution. However, the time complexity of this method is $\mathcal{O}(2^n)$ for an ILP problem with n variables, rendering it unacceptable for large networks. Therefore, we must consider a trade-off between calculation accuracy and time. One possible alternative to solve the ILP problem is to use a linear programming solver, such as Gurobi [36]. However, due to a large number of variables, it still takes a lot of time to solve it directly in a large-scale QKD network. To address these issues, we design a KUR-based algorithm to reduce time complexity, as shown in **Algorithm 1.**

The KUR-based algorithm consists of three main steps, which are elaborated below. Firstly, we find the available path set P_i for the secret-key request i by Dijkstra's algorithm, which can solve the single-source shortest path problem efficiently with non-negative edge weights. Based on the principle of key relaying, a quantum key agreement request consumes an equal amount of key resources on each link of the selected path during key distribution between distant nodes. Considering the uniformity of the QKD devices, each hop takes the same amount of time to relay raw key materials. Therefore, the constraint (4e) can be converted to a constraint on the number of key-relaying hops with the maximum transmission delay and processing delay in the QKD network. Line 3 can ensure the selected paths satisfy the time consumption constraint by limiting the number of hops. Then, we try to relay raw key materials for each key packet r_i^j through idle links. In other words, from lines 5 to 15, for each key packet, all selected paths P_i are evaluated. Among these paths, the one that minimizes the impact on the maximum key utilization ratio in the QKD network is selected as the service path, provided that it meets the on-demand key supply service constraint. This method improves the overall robustness of the QKD network by selecting multiple paths, along with increasing the utilization of key resources, resulting in a decrease in the maximum key utilization ratio. Finally, by repeating the previous two steps, we execute the KUR-based algorithm for each request i in the QKD network to determine the appropriate paths for each key packet. Our proposed algorithm enables us to solve the ILP problem with constraints and obtain the resource allocation method.

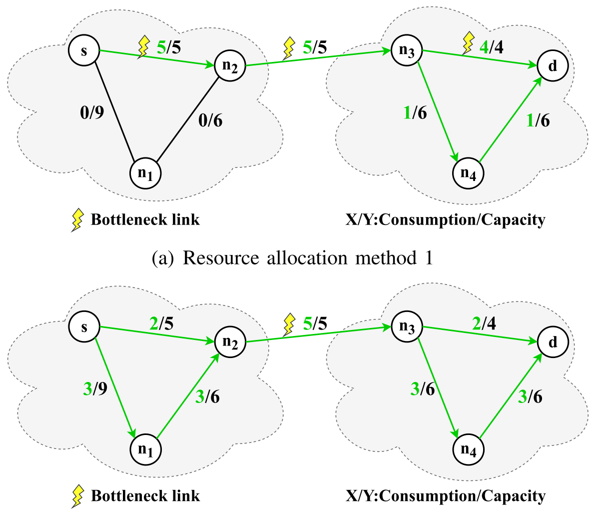
The worst-case time complexity of the KUR-based algorithm is discussed as follows. For request $i \in R$, lines 4 to 16 perform the resource allocation for each key packet. If we use $|A|$ to indicate the number of elements in the set A, the worst-case complexities of line 3 and lines 6 to 12 are $\mathcal{O}(|V|^2)$ and $\mathcal{O}(|P_i|)$, respectively. Thus, for the set of requests R, the overall time complexity of this KUR-based algorithm is $\mathcal{O}(|R| \cdot (|V|^2 + |r_i| \cdot |P_i|))$. In other words, we can solve the resource allocation problem by the KUR-based algorithm in polynomial time complexity.

B. Long-Haul Backbone Networks

The core of the Internet consists of several large networks (often referred to as backbones) [37]. As an add-on technology and service to communication networks [38], long-haul backbone networks play a crucial role in QKD networks, such as integrated space-to-ground QKD systems [9]. Effective load balancing in backbone networks has been demonstrated to enhance throughput and reduce latency in real-world deployments [39], such as inter-datacenter backbone infrastructures [40]. Therefore, achieving load balancing in long-haul backbone networks is crucial.

However, when a large-scale QKD network handles numerous quantum key agreement requests requiring multiple paths to relay the raw key materials simultaneously, the long-haul backbone links can quickly turn into bottleneck links. By definition, α represents the maximum key utilization ratio and indicates whether there is a bottleneck link in the QKD network. Since the optimization objective focuses only on the link with the maximum key utilization ratio, if the bottleneck link is inevitable during the resource allocation, we obtain multiple resource allocation methods by the KUR-based algorithm. Fig. 3 illustrates these resource allocation methods.

As shown in Fig. 3, the amount of keys required for the request $\langle s, d \rangle$ is 5 units. Since the source node s and destination node d are located in different sub-networks, the QKD network must relay the keys across the long-haul backbone link (n_2, n_3), which leads to a maximum resource utilization ratio $\alpha = 1$. In this scenario, the KUR-based algorithm cannot further optimize the resource allocation, which may result in unnecessary bottleneck links within the network, such as edges (s, n_2) and (n_3, d) in Fig. 3(a). In contrast, the optimal resource allocation method is illustrated in Fig. 3(b), where key resources are redistributed from



(a) Resource allocation method 1

(b) Resource allocation method 2

Fig. 3. Different resource allocation methods with inevitable bottleneck links.

unnecessary bottleneck links to other idle links. To achieve this, it is important to filter out inevitable bottleneck links from the QKD network, ensuring load balance across the remaining links. This approach prevents concurrent requests from being impacted by unnecessary bottlenecks.

To achieve efficient and robust resource allocation even in the long-haul backbone networks, we propose the Long-haul Backbone Network (LBN) KUR-based algorithm, as shown in **Algorithm 2**. This algorithm extends the existing KUR-based algorithm by introducing a dynamic bottleneck link filtering mechanism, which identifies and eliminates unnecessary bottleneck links to improve load balancing. Initially, the algorithm detects bottleneck links, defined as links where the key resource utilization ratio reaches its upper limit ($\alpha(u,v) = 1$). These links are temporarily added to the bottleneck link set B. In lines 6 to 18, the algorithm iteratively excludes edges from the bottleneck link set B and reapplies the KUR-based algorithm. If a feasible resource allocation exists without utilizing excluded bottleneck links ($\alpha \neq 1$), those links are deemed unnecessary bottleneck links and are excluded from the set B. However, if excluding a particular edge results in an infeasible allocation (i.e., α remains equal to 1), the algorithm restores it to the bottleneck link set B. After filtering out unnecessary bottleneck links, the algorithm performs the final resource allocation using only the remaining edges in the path set P_i (excluding those in B).

By dynamically identifying and filtering out unnecessary bottleneck links, the LBN KUR-based algorithm ensures that traffic is redistributed away from unnecessary bottleneck links, thereby enhancing load balancing. Compared to the KUR-based algorithm, this algorithm improves the efficiency and robustness of QKD networks, especially in long-haul transmission scenarios.

The worst-case complexity of the LBN KUR-based algorithm is discussed as follows. Lines 8 to 15 iteratively exclude the bottleneck links from the selected paths and apply the KUR-based resource allocation to the remaining links. From

Algorithm 2: LBN KUR-Based Algorithm

Input: $G = (V,E),R,D$;
Output: Assigned $f_i(u, v)$ in terms of allocated links, α, updated stored key resources;

1 Initialize the set of bottleneck links $B \leftarrow \emptyset$;
2 **for** *each secret-key request* $i \in R$ **do**
3   Find the path set P_i with time consumption constraint by Dijkstra's algorithm;
4   Algorithm 1 without the edges in B;
5   **if** $\alpha = 1$ **then**
6   Get the set of bottleneck links B_{temp};
7   **for** *each edge* $(u, v) \in B_{temp} \setminus B$ **do**
8             $B \leftarrow B \cup \{(u,v)\}$;
9   Algorithm 1 without the edges in B;
10   **if** $\alpha \neq 1$ **then**
11   break;
12   **end**
13   **else**
14                 $B \leftarrow B \setminus \{(u,v)\}$;
15   **end**
16   **end**
17         $B \leftarrow B \cup B_{temp}$;
18   Algorithm 1 without the edges in B;
19   **end**
20 **end**

Section IV-A, the time complexity is $\mathcal{O}(|V|^2 + |r_i| \cdot |P_i|)$. During the path selection, the number of hops is constrained by the time consumption constraints. Therefore, for the selected path set P_i, the time complexity of lines 7 to 16 is $\mathcal{O}(|P_i| \cdot (|V|^2 + |r_i| \cdot |P_i|))$. Thus, the overall time complexity of this LBN KUR-based algorithm is $\mathcal{O}(|R||P_i| \cdot (|V|^2 + |r_i| \cdot |P_i|))$. In this case, we can efficiently solve the resource allocation problem within long-haul backbone networks while maintaining polynomial time complexity.

In summary, the LBN KUR-based algorithm is more adaptive and can be applied to achieve load balance in larger-scale long-haul backbone networks. In addition, the KUR-based algorithm can be considered as a special case of the LBN KUR-based algorithm, offering a faster solution in small-scale metropolitan area networks. The adaptability of the proposed heuristic algorithm makes it not only suitable for metropolitan networks but also scalable to large-scale long-haul backbone networks.

V. Experimental Simulations

In this section, we perform extensive simulations to evaluate the performance of the resource allocation method proposed in this paper. In Section V-A, we introduce the evaluation metrics and QKD network topologies selected for the experiments. In Section V-B, to demonstrate the superiority of our method, we compare it against existing resource allocation methods. Firstly, we analyze the maximum key utilization ratio on bottleneck links within different experimental topologies. Following this, we evaluate the time complexity and the effect in different QKD networks, showing that our heuristic algorithm significantly reduces time complexity while maintaining performance. Finally, we validate the dynamics of the links and the dynamics of the quantum key agreement requests in Section V-C, illustrating that our scheme can effectively improve the efficiency and robustness of the network compared to existing schemes.

A. Experimental Setup

Evaluation metrics. To evaluate the performance of our resource allocation method, we consider two evaluation metrics targeting the two features of our method. In a QKD network, the amount of keys requested by users often exhibits randomness, and each quantum key agreement request requires the transmission of multiple key packets. Therefore, the QKD network must make full use of the available key resources to provide on-demand services to users. To measure the ability of the QKD network to serve concurrent quantum key agreement requests, we define the Service Ratio (*SR*) as the ratio of received and sent key packets [25]. Specifically, *SR* can be calculated as follows:

$$SR = \frac{NR_s}{NR}, \tag{5}$$

where NR_s is the number of key packets that can be successfully relayed, and *NR* is the total number of key packets that need to be relayed. A higher *SR* indicates stronger network service capabilities and more efficient utilization of key resources.

In addition, due to the limited key generation rate in current QKD networks and the high requirements for keys from dynamic requests, the key generation and consumption are unbalanced. The bottleneck links on the selected path reduce the possibility of providing services to burst quantum key agreement requests. Therefore, the load should be distributed to the links of the whole network rather than being concentrated on some links. We use the standard deviation to measure the degree of load balance in the QKD network. The Balancing Factor (*BF*) can be expressed as:

$$BF = \sqrt{\frac{1}{n}\sum_{(u,v)}(\alpha(u,v) - \overline{\alpha})^2}, \tag{6}$$

where $\overline{\alpha}$ is the average key utilization ratio, and *n* is the number of edges. *BF* describes the key consumption of each link while providing service for the same quantum key agreement requests. When $BF = 0$, it indicates that the key resources of each link in the QKD network are consumed fairly. That is, the smaller *BF*, the more balanced the key generation and consumption, and the better the robustness of QKD networks.

Topology determination. To verify the effect of the proposed method, we choose two different types of topologies. The first type is the classic network topology represented by a topology dataset called the Internet Topology Zoo [41], which is an ongoing project to collect data network topologies from around the world as experimental topologies. The Internet Topology Zoo provides a diverse set of real-world network topologies, allowing us to analyze resource allocation performance across networks of different scales and structures. The details of the selected networks are presented in Table II. We select topologies with significant differences in scale, ranging from 6 to 38 nodes. This selection enables us to evaluate the impact of different resource allocation methods under varying network conditions, demonstrating the adaptability and scalability of our proposed method. The second network type is the long-haul backbone network represented by an integrated space-to-ground quantum communication network [9]. We selected the metropolitan area networks of Hefei and Jinan as two subnets connected by a long-haul backbone network. The Hefei subnet consists of 11 nodes, while the Jinan subnet contains 50 QKD nodes. The QKD network topology and the key generation rates are configured based on the actual data from the integrated space-to-ground QKD network.

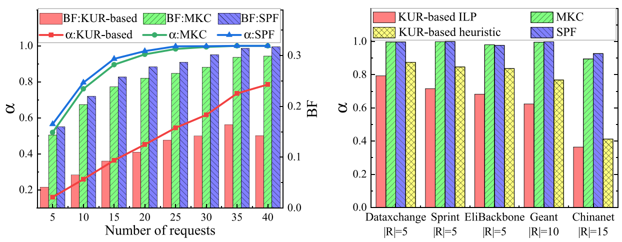
TABLE II
TOPOLOGY INFORMATION

Topologies	Dataxchange	Sprint	EliBackbone	Geant	Chinanet
Number of nodes	6	11	20	37	38
Number of edges	11	18	30	58	62

Comparison methods. To present comprehensive performance comparisons, we choose two comparison methods in our experiments: Shortest Path First (SPF) in the SECOQC QKD network [7] and Minimum Key Consumption (MKC) with a flow-based mathematical model [26]. Next, we introduce them as follows.

- The SPF algorithm, widely adopted in classical networking and extended for QKD networks, utilizes the Dijkstra algorithm to find the shortest available path between the source and destination nodes for each user request. Requests are processed sequentially using a greedy strategy, which may lead to inefficient key utilization, especially when paths become congested or unavailable. If no available path satisfies a request, it is rejected.
- The MKC algorithm takes into account both the quality of service and resource constraints in QKD networks. It formulates the problem using an ILP model that aims to minimize the total key consumption across the entire network. During each time slot, the network gathers all requests and solves the ILP problem to optimally allocate key resources. In cases where key resources are insufficient, requests are rejected. The MKC algorithm can find bottleneck links by considering the network-wide key consumption.

The simulations run on a computer with a 2.7GHz Intel Core i5-11500 CPU and 16GB RAM. The Gurobi optimization software is used to solve the ILP model, while SimQN is installed to implement the heuristic methods. To ensure sufficient statistical accuracy, the simulation results are averaged over 100 repetitions. In the selected metropolitan area network topologies, each node is equipped with the same QKD devices to distribute quantum keys. Specifically, in the SimQN [28] simulation platform, the single-photon emission rate is set to



(a) Chinanet (b) Different topologies

Fig. 4. Effect of different resource allocation methods.

128,000 qubits per second (128 kbps), and the fiber attenuation is configured as 0.2 dB/km. Since the topology data in Internet Topology Zoo contains the latitude and longitude information of the nodes, we can calculate the distance between distant QKD nodes. Combined with the single-photon emission rate, we obtain the key generation rate of each link through simulation. In addition, the quantum key agreement requests are randomly generated, which satisfy the Poisson distribution with parameter λ, and the number of requests in the QKD network is adjusted by continuously adjusting the value of λ. The requests are divided into key packets for transmission, and we fix the packet size to 512 bytes [25] and encrypt data using OTP.

B. Simualtion Results

To evaluate the performance of the resource allocation method proposed in this paper, we first select several metropolitan network topologies to compare the maximum resource utilization ratio and the fairness of the QKD network for different requests. Subsequently, We extend our evaluation to long-haul backbone networks, where the comparison between the LBN KUR-based and the KUR-based algorithms is performed. The focus is on how effectively each algorithm balances the load in the QKD network, especially in the presence of inevitable bottleneck links. Finally, we compare the time complexity of the algorithms in different topologies to evaluate the computational efficiency of each method.

Effect in the classic networks. To evaluate the performance of different resource allocation methods, we first compared the impact of different numbers of requests on the resource allocation strategies using the Chinanet network topology. Furthermore, we also compared the performance of the algorithms under different network scale topologies. The experimental results are shown in Fig. 4.

Fig. 4(a) shows that α increases gradually as the number of quantum key agreement requests increases. When the maximum key utilization ratio α of the QKD network using the SPF and KMC algorithm is close to 1, where a bottleneck link appears, the α of the QKD network using the KUR-based algorithm is just 60%. Since requests are randomly generated, the SPF algorithm leads to a rapid key consumption on bottleneck links since the shortest path is selected. The MKC algorithm can be more efficient than the SPF algorithm since a certain link is selected for multiple requests. However, just like the SPF algorithm, the MKC algorithm does not consider the amount of stored key resources during resource allocation. In contrast, the KUR-based algorithm distributes requests to each link, balancing consumption and generation. Consequently, Fig. 4(a) shows that for the KUR-based algorithm, α increases linearly with the number of requests, with a much lower Balance Factor than the other two algorithms. In other words, our proposed KUR-based algorithm can effectively minimize the maximum key utilization ratio of links in the network while balancing the consumption of keys to reduce the possibility of bottleneck links in the network.

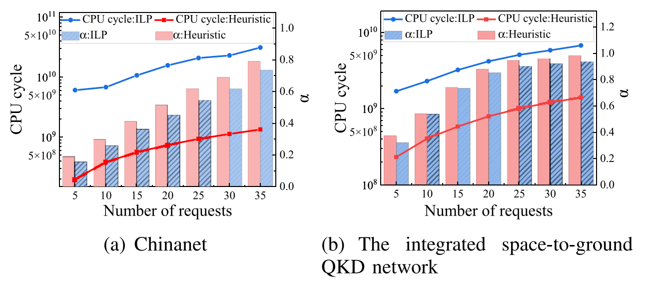
To further verify the performance of different methods under different network topologies, we select five topologies of different scales in Table II. As is shown in Fig. 4(b), the KUR-based algorithm reduces the key utilization ratio by $20\% \sim 50\%$. This difference is due to the scale of the network. The larger the network scale, the more paths the requests can select and the better the algorithm performance.

Effect in long-haul backbone networks. To evaluate the performance of the LBN KUR-based algorithm, we conduct experiments in an integrated space-to-ground quantum communication network, a representative long-haul backbone network. Since the selected network includes two subnets connected by a backbone link, a bottleneck link easily occurs during the key resource allocation process between the two subnets. In the experiments, the quantum key agreement requests are randomly generated, which satisfy the Poisson distribution with strength λ. We vary the request number while keeping the request size fixed at 512 kb, and separately, we evaluate the impact of different request sizes while keeping the average number of generated requests at 10. By controlling these parameters, we analyze how the LBN KUR-based algorithm mitigates bottlenecks and improves key resource allocation efficiency compared to the KUR-based algorithm.

TABLE III
Effect in Long-Haul Backbone Networks

Number of requests	α_{2nd}	α'_{2nd}	$\Delta = \alpha'_{2nd} - \alpha_{2nd}$	Δ/α'_{2nd}
5	0.30460	0.40838	0.09378	25.41%
10	0.52074	0.67998	0.15924	23.42%
15	0.70396	0.84961	0.14565	17.14%
20	0.82874	0.94054	0.11180	11.89%
25	0.88746	0.97884	0.09138	9.34%
30	0.91460	0.99207	0.07747	7.81%
35	0.93659	1.00000	0.06341	6.34%
Average	0.72953	0.83235	0.10283	14.48%
Size of requests (Kb)	α_{2nd}	α'_{2nd}	$\Delta = \alpha'_{2nd} - \alpha_{2nd}$	Δ/α'_{2nd}
256	0.26177	0.36200	0.10023	27.69%
512	0.52074	0.67998	0.15924	23.42%
768	0.73844	0.88714	0.14870	16.76%
1024	0.86156	0.97375	0.11219	11.52%
1280	0.89979	0.99268	0.09289	9.36%
1536	0.93654	0.99648	0.05994	6.02%
Average	0.70314	0.81534	0.11220	15.79%

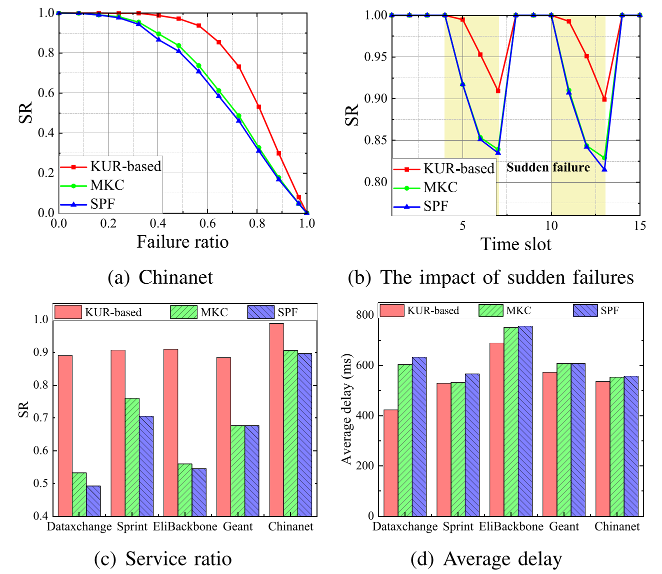
The parameter α_{2nd} is the second highest key utilization ratio with the LBN KUR-based algorithm, and the parameter α'_{2nd} is the second highest key utilization ratio with the KUR-based algorithm. Therefore, $\Delta = \alpha'_{2nd} - \alpha_{2nd}$ and Δ/α'_{2nd} describe the gain in the long-haul backbone networks. As shown in Table III, the LBN KUR-based algorithm can reduce the key utilization ratio by 14.48% and 15.79% on average. When an inevitable bottleneck link occurs, Algorithm 2 distributes the load on the unnecessary bottleneck link to other



(a) Chinanet

(b) The integrated space-to-ground QKD network

Fig. 5. The performance of KUR-based ILP model and heuristic algorithm.



(a) Chinanet

(b) The impact of sudden failures

(c) Service ratio

(d) Average delay

Fig. 6. The effect of the dynamic nature of links.

links, thereby reducing the load on these links and decreasing the number of bottleneck links in the QKD network. Simulation results show that the LBN KUR-based algorithm leads to a more balanced network load in a large-scale long-haul backbone network. This method effectively reduces the load on unnecessary bottleneck links, optimizing the overall performance of the QKD network.

Time complexity. The proposed ILP problem can be solved using linear programming solvers such as Gurobi [36] or by applying the heuristic KUR-based algorithm. In this simulation, we evaluate the performance of these solutions in terms of time complexity under two representative networks: an integrated space-to-ground QKD network and Chinanet.

To compare the time complexity of the different algorithms, we analyze the CPU cycles and present the results in a logarithmic scale for easier comparison. These results are shown in Fig. 5(b) and Fig. 5. From a resource allocation perspective, the ILP model with Gurobi exhibits up to 10% better performance in terms of the maximum key utilization ratio α compared to the heuristic algorithm, as demonstrated in Fig. 5(a) and Fig. 5(b). However, the time complexity of the ILP model with Gurobi grows exponentially, as indicated by the linear increase in the logarithmic coordinates with the increasing number of requests. This exponential growth highlights the inefficiency of solving the ILP problem as the network size expands.

In contrast, the proposed KUR-based heuristic algorithm achieves resource allocation with polynomial time complexity, providing a significant time advantage in larger-scale QKD networks. Notably, the heuristic algorithm can reduce the running time by up to 10 times, with at most a 10% loss in performance compared to the ILP model with Gurobi. As a result, while the ILP model provides better optimality in resource allocation, the heuristic algorithm is more suitable for large-scale QKD network deployments due to its scalability and reduced computational cost.

C. Network Robustness Analysis

To evaluate the impact of resource allocation methods on the robustness of a quantum key distribution (QKD) network, we investigate their performance under different conditions, including link failures and burst quantum key agreement requests. Simulation results show the impact of sudden failures and burst requests under different network topologies. Specifically, we define the service delay for unserved requests as a time-slot length of $D = 1s$ and measure the average delay under both sudden failures and burst requests scenarios.

Dynamic nature of links. In QKD networks, the QKD links may face unforeseen situations that cause the link to fail to relay raw key materials normally. Therefore, we consider that the key relay process cannot be continued when a QKD link is affected. At this time, the SR of the original quantum key agreement requests describes the robustness of the network in the event of link failures. Thus, the dynamic nature of network links significantly affects the robustness and accessibility of the QKD network. The corresponding simulation results are shown in Fig. 6.

Fig. 6(a) shows the relation between the SR and failure ratio for the different algorithms. When links with a substantial accumulation of key resources fail, SR declines significantly for traditional methods like SPF and MKC because there are no alternative routes. However, unlike SPF and MKC which tend to select a single path for key relay, the KUR-based algorithm actively identifies and mitigates bottleneck links by dynamically redistributing key resources across multiple available paths. This method prevents excessive reliance on specific links and effectively reduces the impact of link failures, leading to a service ratio average increase of 16% across varying failure ratios. As shown in Fig. 6(b), when a burst link failure occurs in QKD networks, the remaining links cannot provide on-demand key supply service for the requests, leading to an immediate drop in the SR. However, compared to the SPF and MKC algorithms, the KUR-based algorithm distributes resources more evenly across the network, thus providing more reliable service for requests when the network is unstable.

Furthermore, we evaluate algorithmic performance under different network topologies, as shown in Fig. 6(c) and 6(d). It can be seen that the KUR-based algorithm can significantly improve the SR of requests regardless of the network topology, and SR can be improved by up to 50% for the same 40% failure ratio. Although the proposed method may select longer

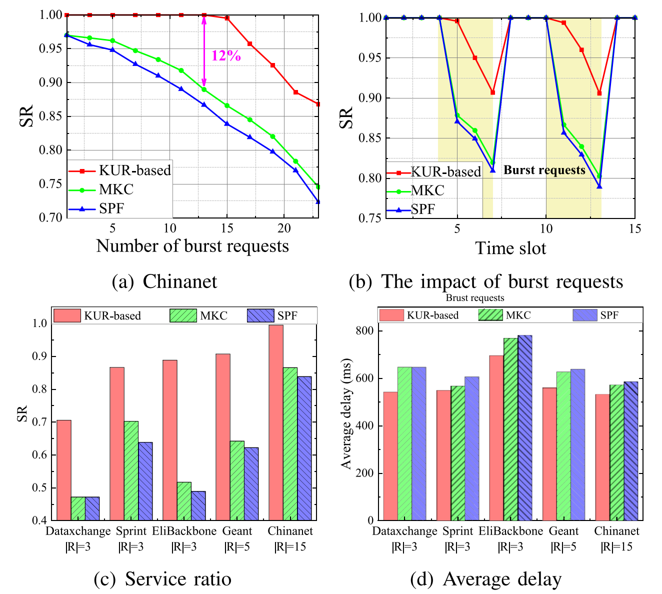


Fig. 7. The effect of the dynamic nature of quantum key agreement requests.

paths for service due to load balancing, it achieves a higher request service ratio, effectively reducing the average delay. This demonstrates the robustness and effectiveness of the KUR-based algorithm in maintaining higher service ratios even under unstable QKD networks.

Dynamic nature of quantum key agreement requests. In QKD networks, due to the dynamic nature of quantum key agreement requests, some burst requests can consume a significant amount of key resources within a short time. These burst requests contribute to bottleneck links, impacting the robustness and accessibility of QKD networks. It is important to ensure reliable service provision for both normal and burst requests. Therefore, the ability to provide service for burst requests can describe the robustness of the QKD network. We evaluate the robustness of our scheme and show simulation results in Fig. 7.

To evaluate the impact of burst quantum key agreement requests in a QKD network, we randomly introduce burst requests three times larger than the normal quantum key agreement requests. Fig. 7(a) illustrates the impact of burst requests on the SPF and KMC algorithms, where links lacking sufficient key resources cannot serve the burst requests. In contrast, the KUR-based algorithm strives to balance stored key resources and ensures link accessibility, significantly enhancing the SR of burst requests. Additionally, the KUR-based algorithm accommodates a greater number of burst requests, considering reliably providing service for them. As illustrated in Fig. 7(b), we evaluate the performance of the proposed method when 15 burst requests are randomly introduced into the QKD network. These burst requests have five times the demand of normal requests and persist for a certain period. The results show that the burst requests negatively impact the SR across all evaluated methods. However, due to the load-balancing capability of our proposed method, it can still maintain a relatively high SR even under such high-load conditions. This demonstrates that our method effectively enhances the robustness of QKD networks by mitigating the impact of burst requests.

Moreover, Fig. 7(c) illustrates that the SR for burst requests is effectively maintained, improving up to 40% due to the KUR-based algorithm. As shown in Fig. 7(d), the KUR-based algorithm dynamically adjusts resource allocation based on real-time key utilization ratios, ensuring that the network can adapt to fluctuations in demand. This method significantly prevents excessive request failures, thereby maintaining a lower average delay even when the network experiences high traffic volatility. The results highlight the proposed algorithm's effectiveness and robustness in managing dynamic and burst requests compared to other algorithms.

VI. Conclusion and Discussion

In this work, we designed an efficient and robust resource allocation method for QKD networks with limited key resources. We first defined the maximum key utilization ratio as a measure of bottleneck links. Then, we formulated an ILP model and presented a heuristic KUR-based algorithm. The proposed KUR-based algorithm was evaluated and compared in different topologies in terms of the maximum key utilization ratio and time complexity. Simulation results prove that the heuristic KUR-based algorithm performs nearly as well as the ILP model but with much lower time complexity. According to the results under different topologies, the KUR-based algorithm reduces the resource consumption of the bottleneck link by up to 50%. Despite the dynamic nature of links and secret-key requests, the KUR-based algorithm can effectively improve the efficiency and robustness of the QKD network.

Due to the layered architecture of QKD networks, the KUR-based algorithm can intricately integrate with existing WDM and TDM methods [42], [43], [44], [45], as demonstrated in real-world QKD networks like the integrated space-to-ground quantum communication network [9]. This integration enhances the capacity of multi-user QKD networks, thereby promoting the extensive deployment of QKD networks on a larger scale [46]. As the number of users in QKD networks grows, the KUR-based algorithm is designed to support QKD networks to provide services in diverse topologies, such as the ring [47], star [48] and mesh [49] structures of short-range, metropolitan and long-haul QKD networks[9]. Notably, since the KUR-based algorithm does not directly interact with physical QKD devices or protocols, it maintains adaptability to various QKD devices and networking equipment from different vendors [7], [8]. However, potential compatibility challenges may arise due to differences in vendor-specific key management interfaces and network control protocols, which could be addressed through standardized interoperability frameworks [38].

The KUR-based algorithm is suitable for resource allocation of a batch of key agreement requests arriving within a time slot in QKD networks, whereas requests often occur in real time. To address this issue, we intend to focus on the resource allocation method for real-time requests. In addition, the numerous secret-key requests on each node result in considerable performance overhead for the centralized controller, which is even more unacceptable in larger-scale networks. Therefore, it can be further considered to perform independent resource allocation on each node to improve scalability.

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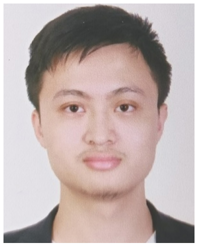

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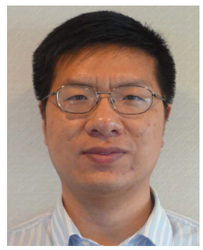

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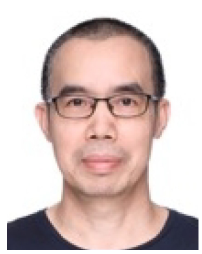

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